

Clearly, a large fraction of the solar spectrum, in particular in the intense 600nm to 1100nm spectral range, is poorly absorbed. This is the reason that conventional wafer-based crystalline silicon solar cells have a much larger thickness, typically 180µm to 300µm. Around forty per cent of the cost of a solar module made from crystalline silicon is the cost of the silicon wafers.

The cost can be reduced if solar cells of reduced thickness deposited on a chip substrate like plastics are used. A major limitation in all thin-film solar cell technologies is that their absorbance of near-bandgap is ineffective, in particular for the indirect-bandgap semiconductor silicon. Therefore structuring the solar cell suitably so that light is trapped inside, in order to increase absorption, is very important. This can be quite effectively done by embedding gold nanoparticles (NPs) on a silicon wafer.

Plasmonic structures can offer at least three ways of reducing the physical thickness of the photovoltaic absorber layer while keeping their optical thickness constant, as shown in Fig. 3.

First, when a gold NP is placed close to the silicon wafer, the light will scatter with an angular spread

in the silicon, which effectively increases the optical path length (Fig. 3(a)). Moreover, the light scattered at an angle beyond the critical angle for reflection, which is sixteen degrees for the silicon/air interphase, will remain trapped in the cell.

Second, gold NPs can be used as sub-wavelength antennas in which the plasmonic near field is coupled with the silicon wafer, increasing its effective absorption cross-section (Fig. 3(b)). Light is trapped by the excitation of localised surface plasmons in gold NPs embedded in the semiconductor. The excited particles' near field causes the creation of electron-hole pairs in the semiconductor.

Third, a corrugated metallic film on the back surface of the thin silicon semiconductor absorber layer provided in SEG can convert sunlight into 'surface plasmon polariton' (SPP) modes, supported at the metal/semiconductor interphase (Fig. 3(c)). SPPs are infrared or visible frequency electromagnetic waves that travel along with a metal-dielectric or metal-air interface. SPP waveguide-coupled back reflector geometry contributes efficient trapping of light and roughly twenty-fold enhancement in the red to infrared solar spectrum.

If the SEG is provided with reflecting metal back contact, light reflected towards the surface will combine with the nanoparticles and partly reradiate into the wafer by the same scattering mechanism (Fig. 4). As a result, the incident light will pass several times through the silicon semiconductor film, increasing the effective path length.

The thickness of the n-silicon semiconductor in earlier SEG is about 170 to 180 micrometres. Using this advanced light trap-

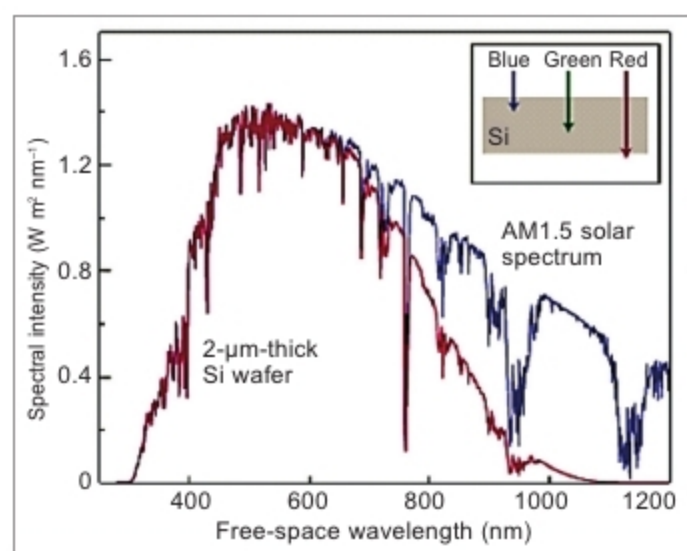


Fig. 2: Absorption of light spectrum in silicon wafer (Credit: HA Atwater-Nature Materials, March 2010)

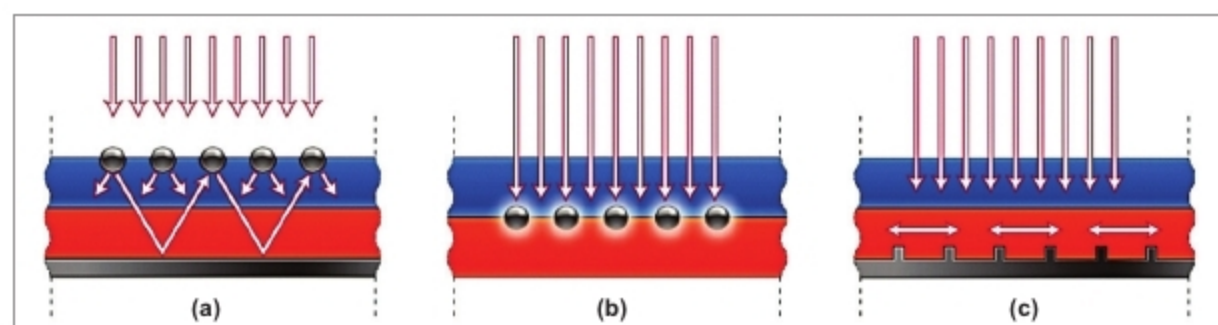


Fig. 3: Three modes of capturing light (Credit: HA Atwater-Nature Materials)

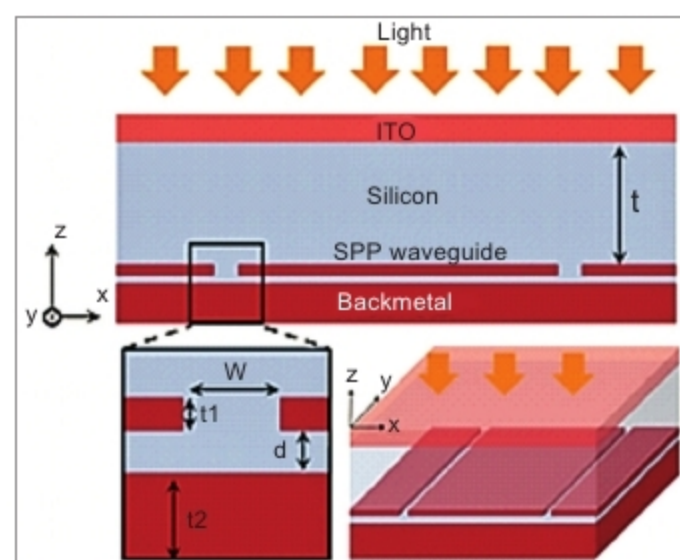


Fig. 4: Schematic diagram of SPP waveguide with reflecting metal back contact

ping strategy along with a properly designed NP architecture, the wafer thickness can be dramatically reduced to around tenth of the aforesaid thickness, keeping solar cell efficiency at twenty per cent. Without a light-trapping technique, thin-film silicon chips would have about seven per cent efficiency. This will reduce the cost of the device greatly.

Thin-film crystalline silicon is formed at low temperature by plasma CVD with submicron grain size. After etching the pattern and providing reflectors, the thin film (~ 17 to 20µm) silicon wafer is fixed to the flexible substrate using epoxy-based adhesive.

## Applications

With lower cost, simplicity, and stability, SEG offers a promising architecture to generate green energy from ambient conditions to power various electronic gadgets and as a part of smart sensor systems, especially in buildings. It can serve as a self-powered sensor for monitoring moving objects by tracking the movement of shadows.

When an object passes by the SEG, it casts an intermittent shadow on the device and triggers the sensor to record the presence and movement of the object. The harvested energy using eight of these generators in the presence of shadows at very low intensity (0.0025 suns) can drive a 1.2V electronic watch. Scientists are looking at ways to embed the small generators into clothing as a smart textile so that we can collect energy on the go.

## Influence of gold nanoparticles

Plasmonic nanoparticles emerged recently as a possible solution to